Who flies with whom?

Identifying group flights in the FFVL archive

Matteo Di Sante | xc-lab meeting, 23 October 2026

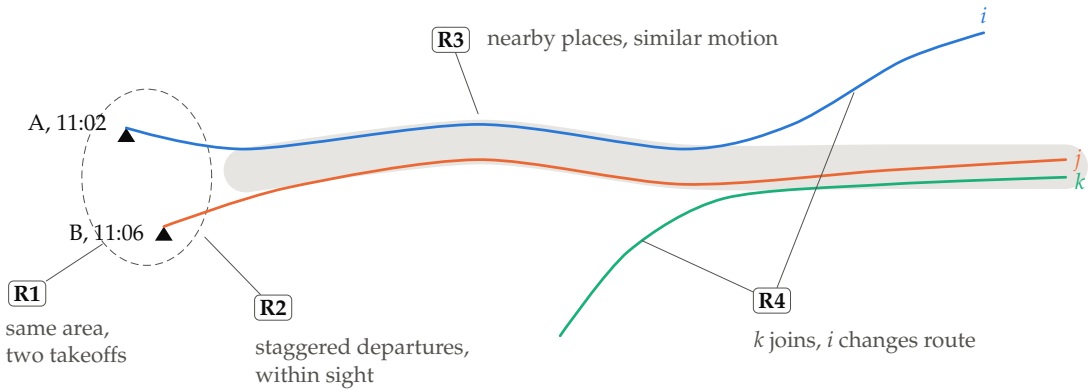
Method note and code: `xc-lab/ideas/group-identification`

Part 1: the idea

What the method does, on pictures

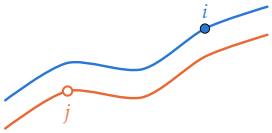
The question

For every flight and every instant: alone or in a group, and with whom?



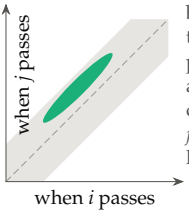
The method in six steps

1 Two tracks



j flies the route of *i*, a little later

2 Same place, with a delay



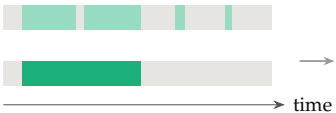
both pass the same place; above the diagonal: *j* passes later

3 Contact



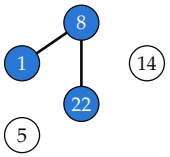
when one of the two pilots is on the path of the other

4 Long contacts



short gaps filled, short pieces dropped

5 Groups at each instant



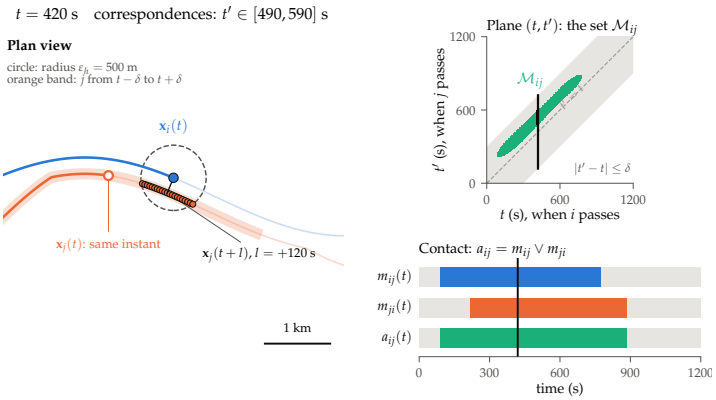
pilots linked by contacts; same members over time: a configuration

6 Episodes



configurations that follow each other without a break

Two pilots on one route, two minutes apart

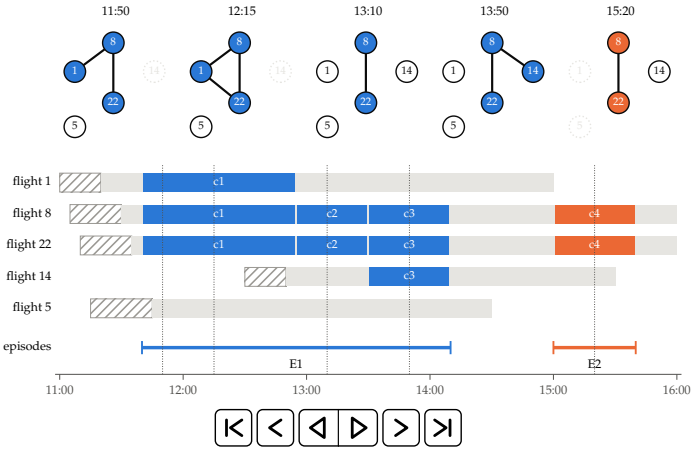


j flies the route of i, 2 min later.

Same instant: **1 km apart**.

Same place, with a delay: **in contact**.

From contacts to groups: an example day



Group: pilots linked by chains of contacts.

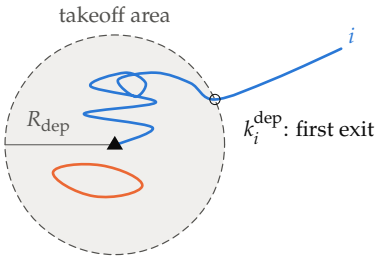
Configuration: a group with the same members $(c_1, \ldots, c_4)$.

Episode: configurations without a break (E1, E2).

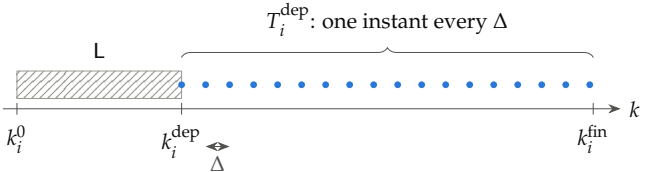
Part 2: the definitions

The same steps, with formulas

Start at the departure



flight j never leaves the disc:
a local flight, all L



Departure.

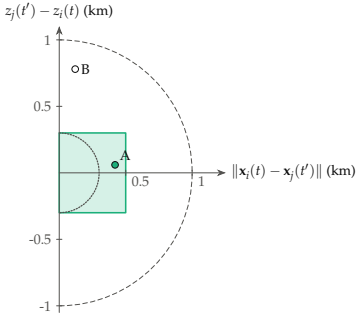
$$k_i^{\text{dep}} = \min\{k : \ \|\mathbf{x}_i(k) - \mathbf{x}_i(k_i^0)\| > R_{\text{dep}}\}, \qquad R_{\text{dep}} = 3 \text{ km}.$$

Before it: local phase L.

Correspondence: the same place at two instants

Correspondence. One clock (UTC) for all flights: i at instant t, j at instant $t' = t + l$, with l the delay:

$$(t, t') \in \mathcal{M}_{ij} \iff |t' - t| \leq \delta,\ \ \|\mathbf{x}_i(t) - \mathbf{x}_j(t')\| < \varepsilon_h,\ \ |z_i(t) - z_j(t')| < \varepsilon_z,\ \ D_{ij}(t, t') = 1$$



dashed: 3D ball, $\varepsilon = 1$ km

dotted: 3D ball, $\varepsilon = 300$ m

box: $\varepsilon_h = 500$ m, $\varepsilon_z = 300$ m

A: glide partner 420 m to the side, missed by the 300 m ball

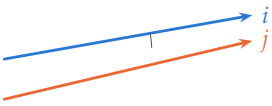
B: 780 m higher, accepted by the 1 km ball

$\delta = 2$ min
$\varepsilon_h = 500$ m
$\varepsilon_z = 300$ m

A flat box: keeps A, rejects B.

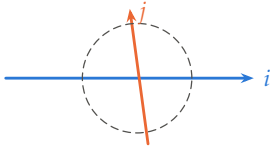
Headings count only when both pilots glide

(a) both glide



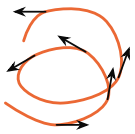
angle $< \theta_{\max}$: ✓

(b) crossing



close, heading elsewhere: ×, excluded

(c) climb or search



the heading turns: no condition

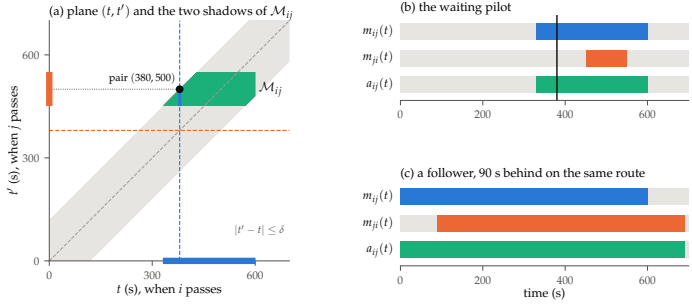
Heading condition.

$$D_{ij}(t,t') = \begin{cases} \mathbf{1}\left[\hat{\mathbf{v}}_i(t) \cdot \hat{\mathbf{v}}_j(t') > \cos\theta_{\max}\right] & \text{if both pilots glide,} \\ 1 & \text{otherwise,} \end{cases} \qquad \theta_{\max} = 45^\circ.$$

Contact: the two shadows of $\mathcal{M}_{ij}$

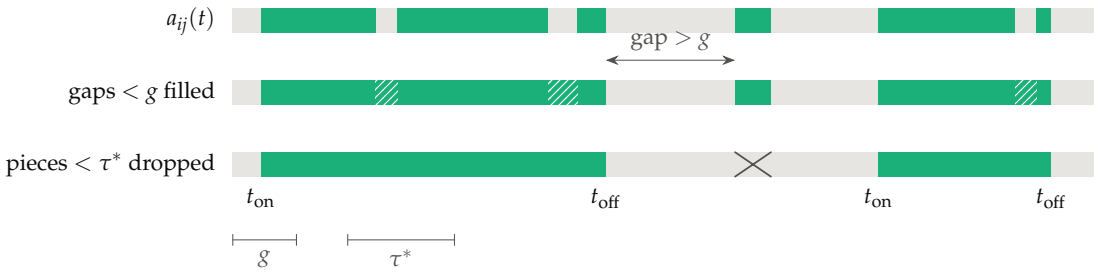
Contact.

$$m_{ij}(t) = \mathbf{1}\big[\exists\, t' : (t,t') \in \mathcal{M}_{ij}\big], \qquad m_{ji}(t') = \mathbf{1}\big[\exists\, t : (t,t') \in \mathcal{M}_{ij}\big], \qquad a_{ij} = m_{ij} \vee m_{ji}$$



(b) i waits in a thermal that j crosses. (c) j follows i 90 s behind.

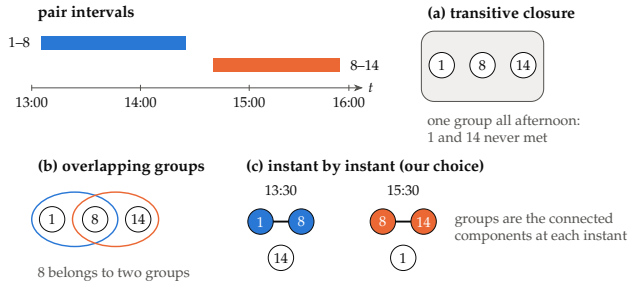
Pair intervals: fill short gaps, drop short contacts



Pair interval. A run of $a_{ij} = 1$ with gaps $< g = 120$ s joined, kept if it lasts $\geq \tau^* = 38.1$ min: $[t_{\text{on}}, t_{\text{off}})$.

τ^*: at most one interval in ten is chance, on pairs shifted in time.

Contact is not transitive: cut the graph instant by instant

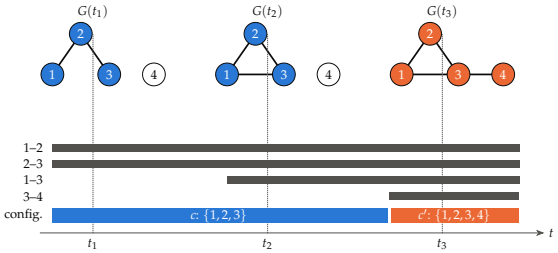


Active graph.

$$G(t) = \big(V(t), E(t)\big), \qquad E(t) = \big\{\{i,j\} : t \in [t_{\text{on}}, t_{\text{off}})\big\}, \qquad V(t) = \text{flights in the air.}$$

A **group** at t: a connected component of $G(t)$ with two or more flights.

Configuration: a group with fixed members

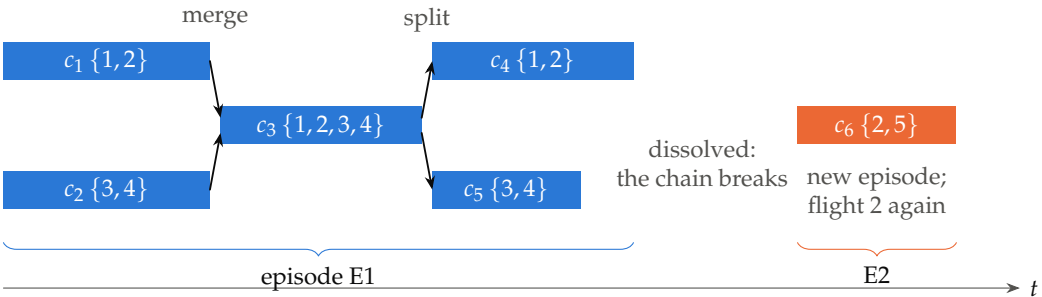


Configuration. $c = (M_c, [t_c^{\text{start}}, t_c^{\text{end}}))$: M_c is a group of $G(t)$ at every t of the interval, and the interval is maximal.

Membership.

$$\gamma_i(t) = \begin{cases} \mathsf{L} & \text{before the departure,} \\ c & i \in M_c,\ t \in [t_c^{\text{start}}, t_c^{\text{end}}), \\ \mathsf{S} & \text{otherwise (solo).} \end{cases}$$

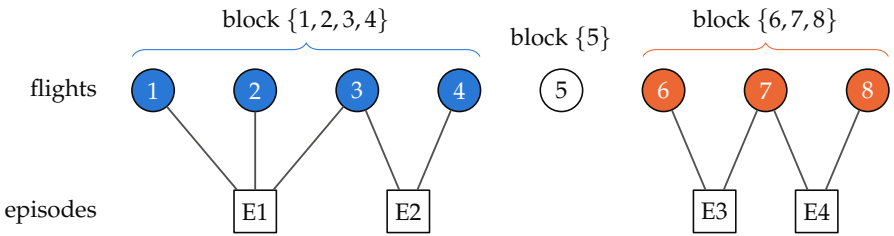
Episode: a chain of configurations



Episode. A connected component of the links

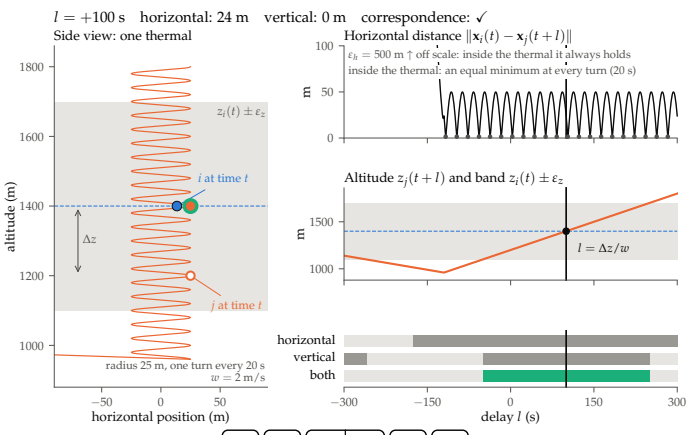
$$c \to c' \iff t_{c'}^{\text{start}} = t_c^{\text{end}} \ \wedge \ M_c \cap M_{c'} \neq \emptyset.$$

Dependency block: the flights tied by episodes



Block. A connected component of the graph flights–episodes of a day. The bootstrap resamples whole blocks.

Who follows whom: a label, not used to build the groups



At instant t, among the delays l with $(t, t+l) \in \mathcal{M}_{ij}$, the one at which j passes closest to where i is (Li et al.):

$$l^*(t) = \arg\min_l \ \|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$$

$\ell_{\text{med}} = \text{median}\, l^*$ over a pair interval; $\ell_{\text{med}} > 0$: j follows i.

Circling: the distance repeats at every turn, so l^* is undetermined inside the window that altitude allows, $\Delta z / w \pm 150$ s at $w = 2$ m/s, against $\delta = 120$ s. Search instants are left out of the median.

On the FFVL archive

21.7%
of paraglider flights fly in a group at some point

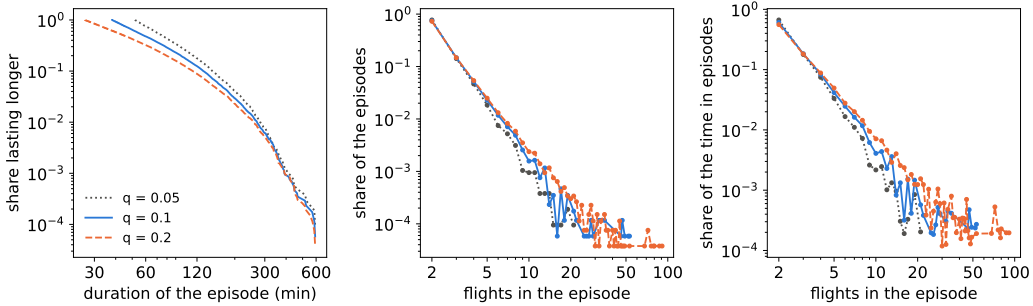
6.9%
of hang-glider flights

17 169
episodes

0.97 h
median episode; 2 flights (median), up to 53

160 637 flights; $\delta = 2$ min, $\varepsilon_h = 500$ m, $\varepsilon_z = 300$ m, $g = 120$ s, $\tau^* = 38.1$ min.

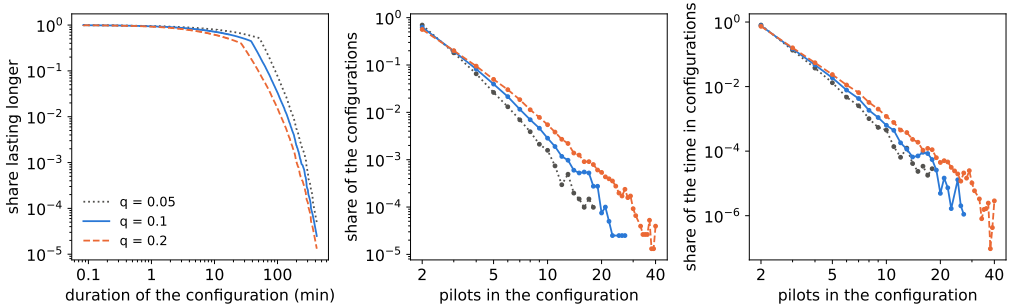
Episodes: how long, how many flights



Whole archive, $\delta = 2$ min. Left: share lasting longer. Middle: share of the episodes of each size. Right: share of the episode time. Size: distinct flights of the episode.

With $q = 0.1$ the median episode lasts 58 min and 74% are pairs. Weighted by time the pairs drop to 61%: large episodes last longer.

Configurations: how long, how many pilots at once



Whole archive, $\delta = 2$ min. A configuration keeps its members; each arrival or departure starts a new one. Size: pilots at once.

Median configuration 30 min, 63% are pairs. Weighted by time the pairs rise to 77%: large configurations change quickly.